AMBA APB

Advanced Peripheral Bus

Theory & Design Reference

Project	AMBA APB Master-Slave RTL Design & Verification
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ARM Reference	AMBA APB Protocol Specification (ARM IHI0011A)
Coverage	APB protocol theory, Master FSM, Interconnect, all 3 Slaves
Purpose	Internship design reference

1. What is AMBA APB?

AMBA (Advanced Microcontroller Bus Architecture) is ARM's family of on-chip interconnect standards. APB — the Advanced Peripheral Bus — is the simplest and lowest-power member of the AMBA family, designed for low-bandwidth peripheral access.

Property	Value
Protocol Family	AMBA (ARM IHI0011A)
Bus Type	Peripheral / Sideband (not main system bus)
Primary Use	UARTs, timers, GPIOs, SPI, I²C, config registers
Clock	Synchronous (PCLK)
Reset	Active-low asynchronous (PRESETn)
Topology	Single Master → N Slaves via Interconnect
Arbitration	None — APB supports only one master
Transfer latency	Minimum 2 cycles (SETUP + ACCESS)
Wait states	Slave-controlled via PREADY (0→N extra cycles)
Error reporting	PSLVERR — slave error flag

🔁 **Analogy:** *APB is like a manager (Master) walking to desks (Slaves) one at a time to deliver or collect files. There is no competition — the manager picks one desk per trip. If the desk isn't ready, it waits (wait states). If the desk doesn't exist, the receptionist (Error Terminator) immediately says 'no such desk'.*

1.1 APB vs Other AMBA Buses

Property	APB	AHB	AXI
Complexity	Low	Medium	High
Bandwidth	Low	High	Very High
Pipeline	No	Yes	Full (5 channels)
Multiple Masters	No	Yes (arbiter)	Yes (per-channel)
Typical Use	Peripheral regs	Main bus	CPU↔Memory, GPU
Power	Lowest	Medium	Highest

2. Signal Definitions

2.1 Clock & Reset

Signal	Direction	Description
PCLK	Input	System clock. All APB signals are sampled on the rising edge.
PRESETn	Input	Active-low asynchronous reset. Forces all outputs and state to zero. 'n' suffix = active-low.

2.2 Master → Slave Signals

Signal	Direction	Width	Description
PSEL	M → S	1 bit	Peripheral Select. Master asserts to select the target slave. Only one slave's PSEL should be high at a time.
PENABLE	M → S	1 bit	Enable. Asserted in the ACCESS phase. APB rule: PENABLE can only go high if PSEL was high in the previous cycle.
PWRITE	M → S	1 bit	Transfer direction. 1 = Write, 0 = Read. Must remain stable from SETUP through ACCESS completion.
PADDR	M → S	8 bits	Address. Identifies which register/memory location to access. Must remain stable during wait states.
PWDATA	M → S	8 bits	Write data. Only relevant on write transfers. Ignored by read-only slaves (ROM).

2.3 Slave → Master Signals

Signal	Direction	Width	Description
PRDATA	S → M	8 bits	Read data from slave. Only valid when PREADY=1 on a read transfer. Ignored on writes.
PREADY	S → M	1 bit	Ready. Slave asserts to complete the ACCESS phase. If 0, master waits (wait state). If 1, transfer completes this cycle.
PSLVERR	S → M	1 bit	Slave Error. Indicates a transfer error. Only meaningful when PREADY=1. Causes are: invalid address, ROM write attempt, unmapped address.

3. APB Transfer Phases

Every APB transfer takes exactly 2 types of clock phases: a mandatory SETUP phase and an ACCESS phase (which may be extended by wait states).

3.1 SETUP Phase

- PSEL is asserted high. PENABLE is kept low.
- Master drives PADDR, PWRITE, and PWDATA (if write) onto the bus.
- The slave samples the address and control signals in this cycle.
- SETUP always lasts exactly 1 clock cycle — the master unconditionally moves to ACCESS.

> 💡 *SETUP is like 'aim'. The master aims at the correct slave and preloads the address. It doesn't fire yet (PENABLE=0).*

3.2 ACCESS Phase

- PENABLE is asserted high. Now both PSEL=1 and PENABLE=1.
- The slave decides whether to accept the transfer now (PREADY=1) or insert wait states (PREADY=0).
- If PREADY=0: master loops in ACCESS. PADDR, PWRITE, PWDATA must remain stable throughout.
- If PREADY=1: transfer completes. Slave delivers PRDATA (on reads) and optionally PSLVERR.

> 💡 *ACCESS is like 'fire and wait for confirmation'. The master fires (PENABLE=1) and waits for the slave to acknowledge (PREADY=1).*

3.3 Transfer Timeline

Signal	Cycle 1 (SETUP)	Cycle 2 (ACCESS)	Wait #1	Wait #2	...	Cycle N+2 (Done)
PSEL	1	1	1	1	...	1
PENABLE	0	1	1	1	...	1
PADDR	valid	stable	stable	stable	...	stable
PWRITE	valid	stable	stable	stable	...	stable
PREADY	—	0	0	0	...	**1 ← DONE**
PRDATA	—	—	—	—	...	valid (read)
PSLVERR	—	—	—	—	...	valid

4. APB Master — 4-State FSM

The APB Master implements a 4-state Mealy/Moore hybrid FSM. It translates testbench/CPU commands (cmd_start, cmd_addr, etc.) into APB protocol signals on the bus.

4.1 State Encoding

State	PSEL	PENABLE	Description
IDLE (00)	0	0	Waiting for cmd_start. Bus is idle. cmd_ready=1.
SETUP (01)	1	0	APB SETUP phase. Driving PADDR/PWRITE/PWDATA. cmd_ready=0.
ACCESS (10)	1	1	APB ACCESS phase. Waiting for PREADY. cmd_ready=1 when PREADY=1.
DONE (11)	0	0	One-cycle cleanup. cmd_done pulsed. Check for back-to-back.

4.2 State Transitions

From → To	Condition	What Happens
IDLE → SETUP	cmd_start=1 AND cmd_ready=1	Master latches cmd_addr, cmd_wdata, cmd_rw into internal registers
SETUP → ACCESS	Always (unconditional)	PSEL=1, PENABLE=1 asserted on bus
ACCESS → ACCESS	PREADY=0	Wait state loop. PADDR/PWRITE stable. Counter increments in slave.
ACCESS → DONE	PREADY=1 AND !cmd_start	Normal completion. cmd_done pulsed. Bus deasserted.
ACCESS → SETUP	PREADY=1 AND cmd_start=1	Back-to-back! Skip DONE. Latch new command immediately.
DONE → IDLE	!cmd_start	Normal end. Bus idle.
DONE → SETUP	cmd_start=1	Another command queued during DONE cycle.
ANY → IDLE	PRESETn=0 (async)	All outputs zeroed immediately, state forced to IDLE.

4.3 Command Interface

The master exposes a simplified command interface to the testbench/CPU, abstracting away all APB protocol details:

Signal	Direction	Description
cmd_addr [7:0]	TB → M	Address of the target register/memory location

Signal	Direction	Description
cmd_wdata [7:0]	TB → M	Write data (only used when cmd_rw=1)
cmd_rw	TB → M	1 = Write, 0 = Read
cmd_start	TB → M	Assert for one cycle to initiate a transfer
cmd_ready	M → TB	High when master can accept a new command (IDLE or ACCESS-with-PREADY)
cmd_done	M → TB	Single-cycle pulse when transfer completes
cmd_rdata [7:0]	M → TB	Read data returned on completion of a read transfer
cmd_error	M → TB	Captured PSLVERR — high if transfer completed with an error

4.4 Back-to-Back Transfers

The master supports pipelined (back-to-back) transfers without any idle cycles between them. This is an important APB optimization:

- When PREADY=1 in ACCESS state, cmd_ready is also asserted combinationally.
- If the testbench asserts cmd_start in the same cycle, the master latches the new command immediately.
- Instead of going to DONE→IDLE, it jumps directly to SETUP for the next transfer.
- This saves 2 clock cycles (DONE + IDLE) per transfer — critical for throughput on zero-wait-state slaves.

> 💡 *Back-to-back is only possible because cmd_ready is combinational, not registered. If it were registered, the testbench would always see it one cycle late and miss the window.*

5. APB Interconnect — Decoder + Response MUX

The Interconnect sits between the single Master and all Slaves. It performs three functions: address decoding, broadcast, and response multiplexing.

5.1 Section 1: Broadcast

Control and data signals (PENABLE, PWRITE, PADDR, PWDATA) are simply fanned out to all slaves identically. Every slave sees every transaction's address and data.

> 🔁 **Analogy:** *A teacher announces homework to the whole class. Everyone hears the same instructions. But only the student whose name is called (PSEL=1) actually does the work.*

5.2 Section 2: Address Decoder

The decoder uses the upper nibble of PADDR to select exactly one slave:

PADDR[7:4]	Slave Selected	Address Range	Module
4'h0 (0x0?)	Slave 0	0x00 – 0x07	apb_slave_ram (N=4)
4'h1 (0x1?)	Slave 1	0x10 – 0x17	apb_slave_reg (N=0)
4'h2 (0x2?)	Slave 2	0x20 – 0x27	apb_slave_rom (N=2)
Anything else	**None**	—	**Error Terminator fires**

Critical: Every slave PSEL is gated with m_PSEL. This means: if the master is idle (m_PSEL=0), all slave PSELs are forced to 0, regardless of address.

> 💡 *Without the m_PSEL gate, the decoder would assert slave PSELs based on address even when no transaction was in progress — a serious protocol violation.*

5.3 Section 3: Response MUX

The MUX routes the selected slave's PRDATA, PREADY, and PSLVERR back to the master:

Condition	m_PRDATA	m_PREADY	m_PSLVERR
!m_PSEL (bus idle)	0	**0**	**0**

Condition	m_PRDATA	m_PREADY	m_PSLVERR
s0_PSEL=1 (RAM)	s0_PRDATA	s0_PREADY	s0_PSLVERR
s1_PSEL=1 (REG)	s1_PRDATA	s1_PREADY	s1_PSLVERR
s2_PSEL=1 (ROM)	s2_PRDATA	s2_PREADY	s2_PSLVERR
m_PSEL=1, no match	0	**1**	**1 — ERROR**

5.4 Error Terminator

If the master tries to access an unmapped address (e.g. 0x35), no slave asserts PSEL. Without the error terminator, the master would loop in ACCESS forever — a deadlock.

> 🔁 **Analogy:** *You dial a phone number that doesn't exist. Instead of ringing forever, the network plays 'this number is not in service' immediately. That's the Error Terminator — an instant response that prevents the master from hanging.*

- Forces m_PREADY=1 — tells the master the 'transfer is complete'.
- Forces m_PSLVERR=1 — tells the master it was an error.
- Master exits ACCESS, pulses cmd_done, and sets cmd_error=1.

6. APB Slaves — Design and Comparison

All three slaves implement the same 3-state FSM (IDLE → SETUP → ACCESS) and differ only in wait-state count, access permissions, and initialisation.

6.1 Slave FSM — Common to All Three

State	Entry Condition	Slave Action
IDLE	PSEL=0 (bus idle)	Wait. Clear PSLVERR so no stale error flags remain.
SETUP	PSEL=1, PENABLE=0	Prepare for transfer. Init wait counter. Abort if PSEL drops.
ACCESS	PSEL=1, PENABLE=1	Count wait states. Assert PREADY when done. Perform R/W. Return to IDLE.

6.2 Wait State Mechanism

The wait_counter counts from 0 to N-1. On each ACCESS cycle where wait_counter < N-1, the slave increments the counter and holds PREADY=0. When wait_counter reaches N-1, it performs the read/write and asserts PREADY=1.

Example for N=4 (RAM):

- Cycle 1: wait_counter=0 → 0 < 3? YES → increment. PREADY=0.
- Cycle 2: wait_counter=1 → 1 < 3? YES → increment. PREADY=0.
- Cycle 3: wait_counter=2 → 2 < 3? YES → increment. PREADY=0.
- Cycle 4: wait_counter=3 → 3 < 3? NO → perform R/W. PREADY=1. Back to IDLE.

> 💡 *N=0 requires special handling: wait_counter < (0-1) = wait_counter < MAX_UINT is always true, causing an infinite loop. The code explicitly checks if N==0 and completes immediately without counting.*

6.3 Slave Comparison

Feature	Slave 0 (RAM)	Slave 1 (REG)	Slave 2 (ROM)
Module	apb_slave_ram	apb_slave_reg	apb_slave_rom
Address Range	0x00 – 0x07	0x10 – 0x17	0x20 – 0x27
Wait States (N)	4	0 (instant)	2
ACCESS cycles	5 (1+N)	1 (instant)	3 (1+N)
Read	✅ Yes	✅ Yes	✅ Yes
Write	✅ Yes	✅ Yes	❌ **Rejected → PSLVERR**
PWDATA port	✅ Yes	✅ Yes	❌ **Not present**

Feature	Slave 0 (RAM)	Slave 1 (REG)	Slave 2 (ROM)
Reset init	All zeros	All zeros	Pre-loaded pattern
PSLVERR sources	Invalid address	Invalid address	Invalid addr + write
Typical real use	SRAM, buffers	Config registers	Boot ROM, LUT

6.4 ROM Pre-loaded Pattern

The ROM is initialised with rom[i] = i × 0x11, giving a distinctive pattern that makes it easy to verify reads visually:

Bus Address	rom[ ]	Value	Calculation
0x20	rom[0]	0x00	0 × 0x11 = 0x00
0x21	rom[1]	0x11	1 × 0x11 = 0x11
0x22	rom[2]	0x22	2 × 0x11 = 0x22
0x23	rom[3]	0x33	3 × 0x11 = 0x33
0x24	rom[4]	0x44	4 × 0x11 = 0x44
0x25	rom[5]	0x55	5 × 0x11 = 0x55
0x26	rom[6]	0x66	6 × 0x11 = 0x66
0x27	rom[7]	0x77	7 × 0x11 = 0x77

7. APB Protocol Rules

These are the binding rules from the ARM APB specification. Violating any of these causes undefined behaviour or bus hangs.

#	Rule	Detail / Why It Matters
1	PENABLE requires prior PSEL	PENABLE must NOT go high unless PSEL was high in the previous cycle. Checked by protocol checker PC-01.
2	PADDR stable in ACCESS	PADDR must not change while PENABLE=1 and PREADY=0 (wait states). Checked by PC-02.
3	PWRITE stable in ACCESS	PWRITE must remain constant throughout an in-progress transfer. Checked by PC-03.
4	PSLVERR only valid with PREADY	PSLVERR is only meaningful when PREADY=1. Checked by PC-04.
5	PRDATA stable in wait	During read ACCESS wait states, PRDATA must not change. Checked by PC-05.
6	cmd_done is single-cycle	cmd_done must pulse exactly once per completed transfer — never held.
7	cmd_ready in IDLE only	cmd_ready is only high in IDLE, or in ACCESS when PREADY=1 (signals back-to-back window).
8	Idle bus: PREADY=0	When m_PSEL=0, the interconnect must NOT assert PREADY or PSLVERR — only during active transfers.
9	Async reset priority	PRESETn has highest priority and overrides all state machine logic immediately (not next clock edge).
10	PSEL abort safe exit	If PSEL drops mid-ACCESS (during wait states), slaves must abort cleanly — no partial writes, no hanging.